

FFR105 HP1 2026

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1 Problem 1.1

We are trying to solve the minimization problem

$$\begin{aligned} & \text{minimize } f(x_1, x_2) = (x_1 - 1)^2 + 2(x_2 - 2)^2, \\ & \text{Subject to} \\ & g(x_1, x_2) = x_1^2 + x_2^2 - 1 \leq 0, \end{aligned}$$

using the penalty method. The penalty function is

$$P(x_1, x_2, \mu) = \mu \max\{x_1^2 + x_2^2 - 1, 0\}^2.$$

The penalized function to minimize is

$$\begin{aligned} f_p(x_1, x_2, \mu) &= (x_1 - 1)^2 + 2(x_2 - 2)^2 + \mu \max\{x_1^2 + x_2^2 - 1, 0\}^2 \\ &= \begin{cases} (x_1 - 1)^2 + 2(x_2 - 2)^2 + \mu(x_1^2 + x_2^2 - 1)^2, & \text{if } x_1^2 + x_2^2 \geq 1, \\ (x_1 - 1)^2 + 2(x_2 - 2)^2, & \text{otherwise.} \end{cases} \end{aligned}$$

As such the gradient becomes

$$\nabla f_p(x_1, x_2, \mu) = \begin{cases} \begin{bmatrix} 2(x_1 - 1) + 4\mu x_1(x_1^2 + x_2^2 - 1) \\ 4(x_2 - 2) + 4\mu x_2(x_1^2 + x_2^2 - 1) \end{bmatrix}, & \text{if } x_1^2 + x_2^2 \geq 1, \\ \begin{bmatrix} 2(x_1 - 1) \\ 4(x_2 - 2) \end{bmatrix}, & \text{otherwise.} \end{cases}$$

Letting $\mu = 0$ we can solve for the unconstrained minimum by solving $\nabla f_p(x_1, x_2, 0) = (0, 0)^T$ which gives us the optima $(x_1, x_2)^T = (1, 2)^T$ which is the minimum since $f(1, 2) = 0$.

Implementing the gradient descent algorithm with step length $\eta = 0.0001$ with convergence criteria $\|\nabla f_p(x_1^{(k)}, x_2^{(k)}, \mu)\| < 10^{-6}$ and starting point $(x_1^{(0)}, x_2^{(0)})^T = (1, 2)^T$ we get the minimizers (x_1^*, x_2^*) shown in table 1.

As μ increases the changes between the minimizers found seem to decrease, indicating convergence. The estimated point which the sequence approaches seems to lie around $(x_1^*, x_2^*)^T \approx (0.31, 0.95)^T$.

μ	x_1^*	x_2^*
1	0.4338	1.2102
10	0.3314	0.9955
100	0.3137	0.9553
1000	0.3118	0.9507

Table 1: Minimizers found using gradient descent for different values of μ .

2 Problem 1.2

Here we are tasked to find the minimum and maximum values of

$$f(x_1, x_2) = 4x_1^2 + 2x_2^3$$

along with the corresponding values of x_1, x_2 over the set $\mathbf{S} = \{(x_1, x_2) \in \mathcal{R}^2 | x_1^2 + x_2^2 \leq 4\}$.

We start by looking for stationary points in the interior of $\mathbf{S}$. Thus we solve

$$\nabla f(x_1, x_2) = \begin{bmatrix} 8x_1 \\ 6x_2^2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

which gives us the trivial solution $P_1 = (0, 0)^T$.

For the boundary $\partial\mathbf{S}$ we use the Lagrange multiplier method. Introducing the Lagrange multiplier λ we get the function

$$\mathcal{L}(x_1, x_2, \lambda) = 4x_1^2 + 2x_2^3 + \lambda(x_1^2 + x_2^2 - 4).$$

Now we want to solve the following equation system

$$\frac{\partial \mathcal{L}}{\partial x_1} = 8x_1 + 2\lambda x_1 = 0, \tag{5a}$$

$$\frac{\partial \mathcal{L}}{\partial x_2} = 6x_2^2 + 2\lambda x_2 = 0, \tag{5b}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = x_1^2 + x_2^2 - 4 = 0. \tag{5c}$$

Equation (5a) gives us two cases, either $x_1 = 0$ or $x_1 \neq 0$. In the first case ($x_1 = 0$) we find from constraint (5c) that $x_2^2 = 4 \iff x_2 = \pm 2$ and we get the stationary points $P_2 = (0, 2)^T, P_3 = (0, -2)^T$. In the second case ($x_1 \neq 0$) we get $\lambda = -4$, which when inserted into (5b) we get $2x_2(3x_2 - 4) = 0$. As such in this case we have either $x_2 = 0$ or $x_2 = \frac{4}{3}$. We continue similarly now for the case of $x_2 = 0$ and get the constraint $x_1^2 = 4 \iff x_1 = \pm 2$ and we get the two stationary points $P_4 = (2, 0)^T, P_5 = (-2, 0)^T$. Finally, if $x_2 = \frac{4}{3}$ the constraint

gives us $x_1^2 + (\frac{4}{3})^2 = 4$, which when solved gives us $x_1 = \pm \frac{\sqrt{20}}{3}$, resulting in the final stationary points $P_6 = (\frac{\sqrt{20}}{3}, \frac{4}{3})^T$, $P_7 = (-\frac{\sqrt{20}}{3}, \frac{4}{3})^T$.

The resulting points and corresponding functions values are found in table 2, where we see that the function takes its minimum at point $(x_1, x_2)^T = (0, -2)^T$ with function value $f(0, -2) = -16$ and its maximum at points $(x_1, x_2)^T \in \{(0, 2)^T, (2, 0)^T, (-2, 0)^T\}$ with function value 16.

Point	$(x_1, x_2)^T$	$f(x_1, x_2)$
P_1	$(0, 0)$	0
P_2	$(0, 2)$	16
P_3	$(0, -2)$	-16
P_4	$(2, 0)$	16
P_5	$(-2, 0)$	16
P_6	$(\frac{2\sqrt{5}}{3}, \frac{4}{3})$	$\frac{368}{27}$
P_7	$(-\frac{2\sqrt{5}}{3}, \frac{4}{3})$	$\frac{368}{27}$

Table 2: Stationary points and corresponding function values.

3 Problem 1.3

We in this problem want to find the minimum of

$$g(x_1, x_2) = (1.5 - x_1 + x_1x_2)^2 + (2.25 - x_1 + x_1x_2^2)^2 + (2.625 - x_1 + x_1x_2^3)^2,$$

with $x_1, x_2 \in [-5, 5]$. In order to do this we will use a stochastic genetic algorithm. We implement the algorithm with the parameters seen in table 3 which, when running the algorithm 10 times with the parameters results in the minimizers (x_1^*, x_2^*) and the corresponding function values found in table 4.

Parameter	Value
Tournament size	2
Tournament probability	0.75
Crossover probability	0.8
Mutation probability	0.02
Number of generations	2000

Table 3: GA parameters used in the optimization.

Run	x_1^*	x_2^*	$g(x_1^*, x_2^*)$
1	2.99999887	0.49999969	2.25×10^{-13}
2	2.99987787	0.49996959	2.39×10^{-9}
3	3.00000423	0.50000118	3.26×10^{-12}
4	2.99999171	0.49999790	1.10×10^{-11}
5	3.00001675	0.50000475	5.31×10^{-11}
6	2.99999231	0.49999820	9.73×10^{-12}
7	3.00000304	0.50000088	1.85×10^{-12}
8	3.00000006	0.49999999	1.49×10^{-14}
9	2.99987787	0.49996959	2.39×10^{-9}
10	3.00004208	0.50001161	3.15×10^{-10}

Table 4: Results from 10 runs of the genetic algorithm.

Running the genetic algorithm for the mutation probabilities (0, 0.005, 0.01, 0.02, 0.05, 0.1) 100 times and collecting the median for each we get the results found in table 5 which is also plotted in plot 1.

The findings show that the mutation probability has a direct effect on the performance of the genetic algorithm. For $p_{mut} = 0$ the median fitness is lower than for the other tested values, indicating that mutation is beneficial for this problem. At mutation probabilities $p_{mut} \in [0.01, 0.05]$ the median fitness becomes ≈ 1 , corresponding to the best fitness score in the test, while at $p_{mut} = 0.1$ the algorithm experiences a slight decrease in median fitness. As such we say that the optimal value of p_{mut} may for this problem lie in the range $[0.01, 0.05]$.

Mutation probability	Median fitness
0	0.992963
0.005	0.999672
0.01	1.000000
0.02	1.000000
0.05	1.000000
0.1	0.999999

Table 5: Median fitness obtained over 100 GA runs for different mutation probabilities.

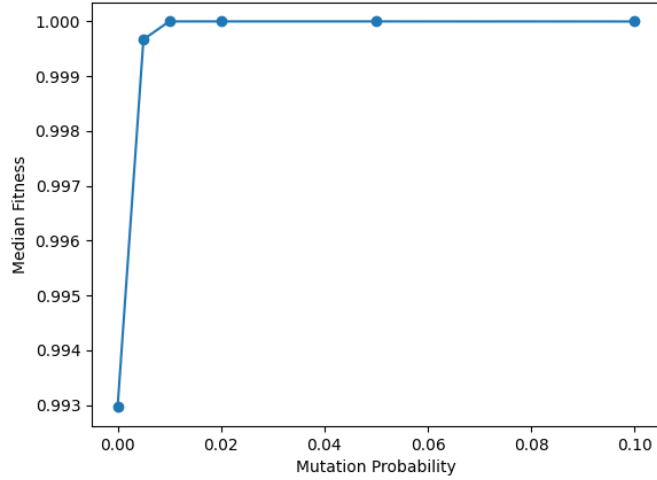


Figure 1: Median fitness of the GA as a function of the mutation probability, based on 100 runs for each value of p_{mut} .

We can from the genetic algorithm results make an educated guess for the optimal $(x_1^*, x_2^*)^T$, which appears to be $(x_1^*, x_2^*)^T = (3, 0.5)^T$. Now, we want to analytically check if the suggested point is a feasible minimum solution to the problem. To do this we compute the gradient of $g(x_1, x_2)$ and check if $(3, 0.5)^T$ is a stationary point.

The gradient of $g(x_1, x_2)$ is given by

$$\nabla g(x_1, x_2) = \begin{bmatrix} 2(1.5 - x_1 + x_1 x_2)(x_2 - 1) + 2(2.25 - x_1 + x_1 x_2^2)(x_2^2 - 1) \\ + 2(2.625 - x_1 + x_1 x_2^3)(x_2^3 - 1) \\ 2x_1(1.5 - x_1 + x_1 x_2) + 4x_1 x_2(2.25 - x_1 + x_1 x_2^2) \\ + 6x_1 x_2^2(2.625 - x_1 + x_1 x_2^3) \end{bmatrix}.$$

Inserting the point $(x_1^*, x_2^*)^T = (3, 0.5)^T$ we find each term gets a factor 0, i.e. that $\nabla g(3, 0.5) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. As such it is a true stationary point to the function $g(x_1, x_2)$ and the minimum since $g(3, 0.5) = 0$ and g is a sum of squares, i.e. non-negative.